Exp 17 MATLAB CODE

```
clc;
clear;
close all;

% Parameters
N = 1000;
SNR = 10;
sps = 20;

% Generate Random Binary Data
data = randi([0 1],1,N);

% BPSK Modulation
signal = 2*data - 1;

% Upsampling
tx = repelem(signal,sps);

% Add AWGN Noise
signal_power = mean(tx.^2);
noise_power = signal_power/(10^(SNR/10));
noise = sqrt(noise_power)*randn(size(tx));
rx = tx + noise;

% Eye Diagram without Noise
figure;
eyediagram(tx(1:2000),2*sps);
```

```
title('Eye Diagram without AWGN');

% Eye Diagram with AWGN
figure;
eyediagram(rx(1:2000),2*sps);
title('Eye Diagram with AWGN');

% Calculate BER
received_symbols = rx(sps/2:sps:end);
detected_data = received_symbols > 0;
detected_data = detected_data(1:N);
errors = sum(data ~= detected_data);
BER = errors/N;
% Display Results
fprintf('SNR = %d dB\n',SNR);
fprintf('Total Bits = %d\n',N);
fprintf('Bit Errors = %d\n',errors);
fprintf('Bit Error Rate (BER) = %.6f\n',BER);
% Plot Transmitted and Received Signals
t = (0:length(tx)-1)/sps;
figure;
subplot(2,1,1);
plot(t(1:400),tx(1:400));
title('Transmitted BPSK Signal');
xlabel('Symbol Periods');
ylabel('Amplitude');
grid on;
subplot(2,1,2);
```

```
plot(t(1:400),rx(1:400));
title('Received Signal with AWGN');
xlabel('Symbol Periods');
ylabel('Amplitude');grid on;
```

